MIL-R-46711D(AT)
24 June 1969
SUPERSEDING
MIL-R-46711C(MO)
9 March 1966

MILITARY SPECIFICATION
RECOVERY VEHICLE, FULL-TRACKED; LIGHT, ARMORED, M578

1. SCOPE

1.1 Scope. This specification covers a front-sprocket-driven, full-tracked, self-propelled, light, armored, recovery vehicle, with individual torsion bar suspension.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

VV-F-800	-	Fuel, Oil, Diesel.

Military

MIL-P-514	-	Plates; Identification, Transportation and Blank.
MIL-L-2104	-	Lubricating Oil, Internal-Combustion Engine Heavy Duty.
MIL-R-3065	-	Rubber, Fabricated Parts.
MIL-L-10295	-	Lubricating Oil, Internal Combustion Engine, Sub-Zero.
MIL-R-46710	-	Recovery Vehicle, Full-Tracked; Light, Armored, M578: Processing for Storage and Shipment of.
MIL-W-46735	-	Winch, Drum: Vehicle Mounting.
MIL-T-46750	-	Transmission, Hydraulic: Model XTG-411-2A Complete; Input Gear Assembly; Main Assembly; Output Gear Assemblies.
MIL-T-46796	-	Engine, Diesel: 8-Cylinder, V-Type 390 H.P.

STANDARDS

Federal

FEDERAL STD. NO. 595 - Colors.

Military

MIL-STD-105	---	Sampling Procedures and Tables for Inspection by Attributes.

FSC-2320

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-193 - Painting Procedures Tactical Vehicles (Tracked and Wheeled).
MIL-STD-210 - Climatic Extremes for Military Equipment.
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642 - Identification Marking Combat, and Tactical Transport Vehicles.

Human Engineering Laboratories (EL)
HEL-S-1-63B - Maximum Noise Level for Army Materiel Command Equipment

DRAWINGS
Engineering List of Drawings (LD)
LD-8736340 - Recovery Vehicle, Full-Tracked; Light Armored; M578

PUBLICATIONS
Engineering Lists of Parts Lists (LP)
LP-8736340 - Recovery Vehicle; Full-Tracked; Light Armored; M578.

Engineering List of Specification (LS)
LS-8736340 - Recovery Vehicle; Full-Tracked; Light Armored; M578.

Master List of Supplementary Quality Assurance Provisions (QM)
QM-8736340 - Recovery Vehicle; Full Tracked; Light Armored; M578.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction vehicle. When specified (see 6.2), a complete preproduction vehicle(s) shall be produced prior to the manufacture or fabrication of vehicles in quantity. The complete vehicle(s) shall be submitted to the Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

for inspection to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling as contemplated under the supply contract.

3.2 First production vehicle. A first production vehicle, shall be fully representative of vehicles proposed to be furnished under the contract, shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2.1 Workmanship and reliability. The workmanship shall be of a quality to assure that the vehicle and components thereof are free from any defects that compromise, limit, or reduce the vehicle or component capabilities herein specified. Vehicle manufactured in accordance with this design shall be subjected to examinations and tests by the Government to ensure vehicle capabilities and performance are in accordance with the following:

(a) Vehicle shall require only organizational maintenance (first and second echelon) during 2,000 miles of normal operation.
(b) Vehicle shall require no replacement or major overhaul of any major automotive component during 4,000 miles of normal operation.
(c) With the exception of track rubber components, all components of complete track assemblies shall require no more than 10 percent replacement during 3000 miles of operation.

3.2.1.1 Reliability operations. To assure that workmanship and reliability requirements are met during the 4,000 miles of operation, the vehicle combat loaded or with a simulated load of equal weight shall be capable of operating as follows:

(a)	Hard surfaced roads	15 percent.
(b)	Hard surfaced roads with rated towed load	5 percent.
(c)	Gravel and dirt roads with 10 percent of this distance under mud conditions	30 percent.
(d)	Gravel and dirt roads with rated towed load with 10 percent of this distance under mud conditions	10 percent.
(e)	Cross country (level and hilly terrain)	35 percent.
(f)	Cross country (level and hilly terrain) with rated towed load	5 percent.

3.2.1.2 Hydraulic system reliability. All hydraulic components shall perform as specified herein without major overhaul or replacement of components. This shall include eight hours of full load operation of each winch (main and hoist).

3.3 Materials.

3.3.1 Qualified products list (QPL). The contractor shall be responsible for using all assemblies and parts necessary in vehicle construction that are

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

listed as qualified products on the QPL and those approved for inclusion on the QPL. Assemblies and parts determined as QPL items appear on the Engineering List of Specifications cited in 3.4. The contractor's manufacturing inspection records shall specifically list all QPL items by part or drawing number(s), name of supplier(s), and number and date of the QPL. When assemblies and parts are approved as qualified products but not yet listed on the QPL, the contractor shall list the products by number and date of the approval document with name of supplier(s).

3.3.2 Ozone resistant. Unless otherwise specified, all rubber products procured under requirements of MIL-R-3065 are to be ozone resistant to the degree specified herein. Rubber components controlled by detail specifications shall be ozone resistant only when required by the specification referenced on the drawing.

3.4 Construction. All vehicles, sub-assemblies, and assemblies shall be fabricated and assembled into a complete vehicle in accordance with drawings, parts lists, and specifications as specified on LD-8736340, LP-8736340, LS-8736340 for the M578, and associated documents and publications referenced therein. All parts requiring identification shall be marked in accordance with MIL-STD-130.

3.4.1 Sealing.

3.4.1.1 Sealer. The sealer used to seal vehicle shall prevent the entrance of water into the hull when vehicle is fording streams or rivers during cross country operation and when operating in deep water (see 3.6.8).

3.4.1.2 Seals. Seals shall prevent the entrance of all foreign matter into the bearings, when fording shallow or deep water. All seals shall prevent the leaking of lubricants from the bearings.

3.4.1.3 Hatch and door seals. When hatches and doors are closed the seals shall prevent the entrance of water into the occupied portions of the vehicle.

3.4.2 Winterization kit. When specified and installed (see 6.2), the winterization kit shall ensure engine starting and operation in ambient temperature of minus 25°F to minus 65°F, within 20 hours of previous run.

3.4.3 Exhaust system. The exhaust system shall prevent the accumulation of carbon monoxide in the occupied portions of the vehicle in excess of 0.01 percent under normal condition of vehicle operation.

3.4.4 Air cleaners. With a maximum air flow of 615 cubic feet per minute through each air cleaner, the cleaners shall remove not less than 98 percent of dust and foreing material, 1 micron in diameter or larger from engine intake air under all vehicle operating conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

3.4.5 Fuel system. During engine operating, the fuel system shall maintain fuel supply to the engine when standing on, ascending, and descending 60 percent grades in forward and reverse gear, and when vehicle is standing on or being operated on 30 percent side slopes, with each side of vehicle up slope.

3.4.5.1 Fuel tank. Provisions shall be made to ensure the internal cleanliness of the fuel tank. All weld slag and foreign material shall be removed prior to initial filling of fuel tank. The fuel tank and all connections shall be free from leakage during all conditions of vehicle operation. The fuel tank shall be capable of receiving 50 gallons of fuel per minute.

3.4.5.2 Fuel and hydraulic lines. Provisions shall be made to ensure internal cleanliness of fuel and hydraulic lines and connections prior to initial servicing of vehicle. All fuel and hydraulic line connections shall be routed to ensure no interference, sharp bends, chaffing, and leakage during all conditions of vehicle operations.

3.4.6 Hydraulic reservoir. Provisions shall be made to ensure the internal cleanliness of the hydraulic reservoir. All weld slag and foreign material shall be removed prior to initial filling of reservoir. The reservoir and all connections shall be free from leakage during all conditions of vehicle operation.

3.4.7 Suspension lockout. The lockout cylinder shall lock the suspension system providing a firm base for winching or hoisting the rated loads.

3.4.8 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation, under all vehicle operating conditions.

3.4.9 Adjustment mechanisms. All adjustment mechanisms shall function properly and maintain adjustment during all vehicle operating conditions.

3.4.10 Fasteners and locks. All fastening and locking devices shall be positive in action. Means for locking vehicle from outside shall be provided.

3.4.11 Stowed equipment. All on vehicle equipment (OVE) items shall be stowed (combat loaded) on the vehicle in the spaces provided.

3.4.12 Engine air induction system. Engine air induction system as fabricated and installed from air cleaner element mounting face to engine turbo-charger intake shall be free of all weld slag. The system shall prevent the entrance of all foreign material, and withstand pressure between 3 and 5 PSI.

3.4.13 Throttle linkage. With throttle linkage attached to engine and throttle pedal fully depressed, the throttle shall be at full rack position.

3.4.14 Fuel shut-off valve. With the engine operating at not more than

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

1000 RPM the engine shall stop within one minute when the manual fuel shut-off valve is placed in the "OFF" position.

3.4.15 Seating. All sliding parts and operating contact surfaces shall be free of paint.

3.4.16 Electrical system.

3.4.16.1 Generator voltage. The engine shall be operated at a minimum of 1,000 revolutions per minute (rpm) for 30 minutes before setting voltage regulator. The vehicle generator output voltage shall be between 27.5 and 29 volts, under all conditions of vehicle operation.

3.4.16.2 Lights. The internal and external lights shall operate as specified through all vehicle operating conditions. All electrical contacts and connections shall maintain positive contact under all vehicle operating conditions. The electrical harness shall be routed and secured to assure that all circuits shall be free from chaffing, abrasive action, and damage from any other source during all conditions of vehicle operation.

3.4.16.3 Slip ring (hull to cab) and all communication circuits. The communication electrical circuits shall be capable of delivering 24 volts nominal to the communication equipment. The slip ring ground and communication circuits shall maintain continuity throughout continuous 360 degrees of cab traverse in either direction without evidence of shorts. Each circuit shall maintain continuity from end to end without shorts from pin to pin or from the pins to the connector shell.

3.4.17 Vision devices and receptacles. Provisions shall be made to ensure that receptacles incorporated in the vehicle will accept applicable vision devices without binding or interference. When installed into the receptacles, vision devices and seals shall prevent the entrance of water into vehicle.

3.4.18 Welding repairs. Welding repairs of any type or class shall be made only when and to the extent, specifically authorized by the procuring activity.

3.5 Break-in run.

3.5.1 Preparation. Prior to break-in run, except for equipment and component installation and servicing performed by the contractor as a final processing of the vehicle for acceptance, the vehicle shall be completely assembled and all adjustments completed. Vehicle, including all components requiring lubrication shall be lubricated, and serviced with diesel fuel conforming to grade DF 2 of VV-F-800. Lubricants and fluids shall be as specified in production lburication Drawing C10910925, listed on LD-8736340. Specific applications by grades or types for various expected seasonal temperature ranges shall be as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

3.5.2 Oil pressure and temperature. Prior to the start of the break-in run, the engine shall be operated at 750 to 1000 rpm until lubricating oil is at operating pressure and temperature. If at any time during the break-in run the engine has been stopped 30 minutes or more, the engine shall again be operated as above, before continuing break-in run. Proper oil pressure and temperature shall be maintained during operating on level ground, 60 percent grades and 30 percent side slope.

3.5.3 Distance operated. Prior to tests, each vehicle shall be given a break-in run, for the distances specified in Table I on smooth, level, hard-surfaced roads. The vehicle shall be operated in both directions over the undulating section of road (see figure 1) during "a" and "b" divisions of the break-in run.

Table I. Speeds and distances for break-in run

Division of run	Speed mph	Distance miles	Test conditions
a	0 to 10	10	Track (road)
b	11 to 20	25	Track (road)
c	21 to max.	5	Track (road)

3.5.3.1 Reverse operation. After each division of the break-in run, the vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet, or simulated operation in skid-type dynamometer at speeds between 5 and 15 mph.

3.5.4 Simulated break-in run. Prior to tests, each vehicle may be given a simulated break-in run for distances specified in Table II for a skid type dynamometer and on smooth, level, hard-surface roads in lieu of break-in run specified in 3.5.3. The vehicle shall be operated over an undulating section of road in accordance with figure 1 during "d" division of the simulated break-in run.

Table II. Simulated break-in run

Division of run	Speed mph	Miles of test	Test conditions
a	0 to 10	15	Dynamometer
b	11 to 20	15	Dynamometer
c	21 to max.	5	Dynamometer
d	11 to 20	10	Track (road)

3.5.5 Conditions after run. After completion of the break-in or simulated break-in run the engine shall idle smoothly. Prior to vehicle being submitted

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

to the Government for acceptance, all suspension wheel bearings shall be checked to determine if adjustments are correct and readjusted as required. There shall be no defects indicative of damaged components, or maladjustments that may cause faulty vehicle operation.

3.6 Performance. A complete vehicle, combat loaded (see 3.4.11) and serviced shall perform as specified herein. Vehicle, serviced, (see 3.5.1) and equipped for existing climatic conditions, shall operate as specified without special equipment. Vehicle shall operate on 60 percent longitudinal grades, headed up and headed down gredes, and on right and left 30 percent side slopes, with each side of vehicle up slope. Performance shall be demonstrated on smooth, level, hard-surfaced roads free of loose material, except a section of road shall be as shown in figure 1.

3.6.1 Environmental. The vehicle shall operate in ambient air temperature of plus 115° to minus 25°F, and with special equipment installed, at temperature ranges of minus 25° to minus 65°F. The complete vehicle when in storage shall withstand climatic extremes as specified in MIL-STD-210, without deterioration that may cause failure of any component part of the vehicle.

3.6.2 Power package and train. The power package when installed into vehicle shall function throughout all gear and speed ranges without loss of lubricants, and damage that may cause failure of vehicle. The power train and associated controls shall operate throughout all speed and steering ranges without binding of linkages, loss of lubricants, grabbing, and chattering or slippage when controls are applied to control, stop, and hold vehicle.

3.6.2.1 Engine. Engine conforming to MIL-E-46796, when installed into vehicle it shall meet all performance requirements specified herein.

3.6.2.1.1 Cooling system. The engine water temperature, engine and transmission lubricant temperature, shall not exceed values shown in Table III when the vehicle is operated under any one or a combination of the following conditions:

(a) Ambient temperature from 0°F to 115°F.
(b) Under full load.
(c) At highest available torque output.
(d) At all speeds including a minimum speed of 2 1/2 mph in low hear with no track slippage when operating vehicle on smooth, level, hard-surfaced roads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

Table III. Operating temperatures

Maximum Water Temp. from engine	Maximum Engine Oil Temp. in sump	Maximum transmission oil temp. into cooler
230°F	275°F	300°F

3.6.2.2 Transmission. Transmission conforming to MIL-T-46750, when installed, into vehicle it shall meet all performance requirements.

3.6.3 Speeds.

3.6.3.1 Speeds no load. Vehicle shall operate at a sustained speed of 30 mph on relatively level ground and shall maintain a minimum of 1.5 mph in lowest gear without towed load. The vehicle shall attain a maximum speed of 34 mph. With vehicle traveling between 20 mph and maximum speeds, the drift from a straight line shall be not more than 3 feet in 100 feet.

3.6.3.2 Speeds with towed load. The vehicle on relatively level ground shall tow a 25 ton, track laying free rolling vehicle at a maximum speed of not less than 20 mph and shall maintain a speed of 15 mph. The vehicle shall also attain a speed of 2 1/2 mph with towed load on relatively level and hilly cross country terrain.

3.6.4 Acceleration. Vehicle shall accelerate from a standing start to 20 mph in not more than 25 seconds.

3.6.5 Engine starting on grades and slopes. When standing on a 60 percent grade, for not less than 2 minutes, with engine operating under no load between 750 to 1000 rpm, the engine shall be stopped for not less than 2 minutes. The engine shall restart in not more than 1 minute when headed up and headed down grade. Similar engine stopping and starting shall be accomplished on 30 percent right and left side slopes.

3.6.6 Braking.

3.6.6.1 Stopping. Vehicle traveling at a speed of 20 mph shall stop within a distance of 35 feet from point of brake application, drift not to exceed 3 feet.

3.6.6.2 Holding. With vehicle combat loaded or with a simulated load of equal weight and standing on a 60 percent grade, with service brakes applied, the vehicle shall be held stationary when headed up and headed down grade. With parking brake engaged and all other holding devices inoperative, the vehicle shall be held stationary when headed up and headed down grade.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

3.6.7 Turning. The vehicle shall make continuous, full 360 degree turns to the right and to the left when operating on smooth, level, hard-surfaced terrain. When gear steer is employed, the rear point of the outer edge of the vehicle shall be within a 100 foot diameter circle. When clutch brake steer is employed, the vehicle's track pattern shall be within a 35 foot diameter circle.

3.6.8 Fording shallow water. Vehicle shall ford hard-bottom salt or fresh water crossing of 42 inches in depth, including wave height without special equipment. With vehicle relatively level and operating in water up to 42 inches in depth for 15 minutes, the average accumulation of water shall be not more than 1 inch in bottom of the hull.

3.6.8.1 Front and rear drain valves. Drain valves shall oper and close without indications of binding, sticking, and the controls shall be free from binding and interference. The drain valve controls shall lock valve in an open position.

3.6.9 Climbing. Vehicle shall climb lingitudinal grades of 60 percent in forward and reverse gear without stalling or damage to power plant and power train.

3.6.10 Trench crossing. Vehicle shall cross trenches 36 inches in depth and 75 inches in width without stalling or damage to vehicle and components.

3.6.11 Vertical obstacle. Vehicle, with fender extensions installed shall cross vertical obstacles up to 34 inches in height without damage to vehicle and components. Vehicle without fender extension shall cross vertical obstacles up to 40 inches in height without damage to vehicle and components.

3.6.12 Hydraulic system. With the vehicle engine operating at governed speed of between 1200 rpm minimum with full load and 1375 ± 25 rpm no load, the hydraulic system shall operate with smooth positive control of the following: Boom, spade, main winch, hoist winch, traverse mechansims, winch and cab traversing brakes, impact wrench, suspension lockout, and level wind device throughout their entire operating range under all operational conditions specified herein. The hydraulic system shall not be operated when the oil in the hydraulic reservoir exceeds 170°F. The hydraulic system shall operate satisfactorily using oil conforming to MIL-L-10295 (sub-zero) or oil conforming to MIL-L-2104, as required in high ambient air temperatures.

3.6.12.1 Descent rate of boom. With the hydraulic system stabilized, oil temperature between 70°F and 90°F, boom in a horizontal position and loaded with 10,000 pounds, the settling (descent) shall be not more than 3/4 inch in a 6 minute time period.

3.6.12.2 Boom. The hydraulically operated boom without load shall be hydraulically raised from its travel position to its maximum height in not more than 30

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

seconds. From this position, it shall return to the travel position in not more than 30 seconds. With the vehicle combat loaded, on firm ground having less than 4 percent grade, lockout cylinders locked out, and boom extended to its maximum length from the rear of vehicle and:

(a) With the spade emplaced, the vehicle shall remain stabilized, the hydraulically actuated boom fully extended shall hoist 30,000 pounds to the maximum height position and return. With the boom in a horizontal position, the boom cylinder relief valve shall limit the boom's vertical lift capacity to 34,000 pounds.

(b) With the spade in its stowed position, the boom shall lift 15,000 pounds and hold the load clear of the ground while the vehicle is driven forward for not less than one vehicle length.

3.6.12.3 Winches. Winches conforming to MIL-W-46735, when installed into vehicle winches shall meet the requirements specified herein.

3.6.12.4 Main winch. The main winch in low gear shall have a line pull of 60,000 pounds minimum on bare drum. The hydraulic relief valve shall limit the line pull to 63,000 pounds maximum on bare drum. With the winch under 40,000 pounds load, the spooling rate shall be not less than 16 feet per minute on bare drum.

3.6.12.5 Hoist winch. With the fourth layer of cable on the winch drum, and winch in low gear using a 2 part line it shall have a lifting capacity of 30,000 pounds. With the fourth layer of cable on the winch drum, the hydraulic relief valve shall limit the lift to 34,000 pounds. With the fourth layer of cable on the winch drum and with a 30,000 pounds load and using a 2 part line the spooling rate shall be not less than 25 feet per minute.

3.6.12.6 Winch brakes. The automatic mechanical brakes on both winches shall stop and hold the line pull loads specified in 3.6.12.3 and 3.6.12.4, and prevent cable from unwinding after hydraulic power has been removed. During winch operations, the brake shall release without dragging or shattering.

3.6.12.7 Spade. The spade shall stabilize the vehicle when winching or hoisting maximum loads specified in 3.6.12.3 and 3.6.12.4 from an emplaced position, the spade hoisting system shall raise the spade to its stowed position in not more than 4 seconds.

3.6.12.8 Traversing. The cab shall traverse under either of the following conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

(a) The vehicle combat loaded, on firm ground having a longitudinal grade of 10 percent, spade in stowed position, lockout cylinders locked out, boom at maximum reach and loaded with 15,000 pounds, the cab shall traverse 1 full revolution per minute (rpm). A guide line shall be attached to the load.

(b) The vehicle, combat loaded, on firm ground having a 7 percant side slope, spade is stowed position, lockout cylinders lockout, boom at minimum reach and loaded with 25,000 pounds, the cab shall traverse 90 degrees left and right from center line of vehicle at the rate of 1 rpm. A guide line shall be attached to the load.

3.6.12.8.1 Traversing brakes. With the vehicle positioned and loaded as specified herein, and the cab at any position, the traversing brake shall be capable of preventing cab rotation.

3.6.12.8.2 Emergency hydraulic pump. A manually operated hydraulic pump shall be capable of supplying adequate oil pressure for releasing tow winch, boom winch, traverse motor brakes, and for raising the spade under emergency conditions.

3.6.12.8.3 Level wind device. When the tow winch level wind device senses fleet angle deviations it shall activate the cab traverse system to automatically correct the fleet angle by alining cab with angle of tow winch cable. When the tow winch is to be used for special operation, the level wind device shall be capable of being switched off. When cab is traverse automatically, the manual operated foot pedal control shall override the level-wind automatic traverse control device.

3.7 Radio suppression. Unless otherwise specified (see 6.2) the complete vehicle shall be radio interference suppressed in accordance with tactical vehicle requirements of Class IIIA of MIL-STD-461.

3.8 Painting, marking and data plates.

3.8.1 Painting. Unless otherwise specified (see 6.2), the exterior and interior of turret, components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with MIL-STD-193. Vehicle exterior paint and non-skid paints shall match color chip X24087 (olive drab) Fed. Std. No. 595.

3.8.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-642.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

3.8.3 Name, shipping, and service data plates. Data plates shall be in accordance with applicable drawings or as specified by the procuring activity. The plates shall meet the performance requirements of MIL-P-514.

3.9 Vibration and noise level.

3.9.1 Equipment operation. Equipment operation for training or maintenance purposes shall not exceed the maximum acceptable noise level. Noise reduction principles and techniques applied to the equipment shall be determined by the procuring activity. Should the cost of reducing the sound pressure levels to meet the maximum of or below the requirements of Table IV become prohibitive, it shall be necessary to obtain the approval of the procuring activity before preceeding with the reduction.

3.9.2 System effectiveness. The reduction of the sound pressure level to or below the maximum acceptable noise level shall not degrade the system effectiveness.

3.9.3 Crew positions. The noise level at crew positions shall not exceed the limits specified in HEL S-1-63B.

3.10 Fire extinguisher. The fixed fire extinguisher system, when installed in accordance with applicable drawings shall be readily accessible for servicing and operation internally and externally. In ambient of 72°F the maximum effort required to discharge the system, either internally or externally, shall be not more than 35 pounds.

3.11 Workmanship. Workmanship shall be of a quality to assure the vehicle and components free of defects resulting from improper manufacturing or assembly practices and shall conform to requirement herein specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

inspection and quality control system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent limitation is included in the approval quality control procedure. The contractor shall notify the Government and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government production inspection measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance product to prescribed criteria has been demonstrated.

4.1.3 Parts and assemblies. Parts and assemblies shall be inspected in accordance with the drawings listed on LD-8736340 and QM-8736340. Items requiring ozone resistant shall have met the "C_1" test requirements of MIL-STD-417.

4.1.4 Qualified products. The contractor's inspection records shall be checked to determine contractors conformance to 3.3.1.

4.2 Preproduction and first production inspection.

4.2.1 Preproduction inspection. A preproduction vehicle (see 3.1) when furnished shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed by the Government at a location approved by the Government. Failure of the preproduction model to comply with any of the requirements specified shall be cause for refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.2.2 First production inspection.

4.2.2.1 Special process inspection. A special process inspection will be condicted by the Government representatives during fabrication of the first production vehicle (see 3.2) to evaluate conformance of materials and workmanship with drawing and specification requirements.

4.2.2.1.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection systems and plan of quality control, will be con-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

ducted on the following: material, welding, surface hardening, heat treating, protective finishes, etc.

4.2.2.1.2 Hull inspection. The hyll shall be inspected prior to covering the weldments with paint or other pertinent components or assemblies.

4.2.2.2 Complete vehicle inspection. The first vehicle when complete, shall be inspected by the Government subsequent to contractor's inspection at the contractor's plant to determine conformance to contract and specification requirements.

4.2.2.2.1 Corrections. Corrections found necessary as a result of above inspection shall be made by the contractor on the vehicle.

4.2.2.3 Production vehicle test. After completion of the first vehicle inspection, a second vehicle shall be selected from the first months production or one of the first ten shall be subjected to examination (see 4.4.2) and tests as specified in 4.5 (a) and (b). Subsequent to examination and tests, the selected vehicle shall be subjected to the 4,000 mile test specified in Table IV. These tests shall be performed by the Government, at a site approved by the Government, which will require a maximum of 90 days. Delays in test completion caused by vehicle breakdown or failure of the contractor to adequately support the vehicle with parts during the test shall not be the basis for adjustment of the contract, delivery schedule or contract price.

Table IV. 4000 Mile Test (Government Proving Grounds)

Course	Mileage & speeds
Hard surface roads	600 miles at varying speeds up to 34 mph combat loaded.
Hard surface roads	200 miles at varying speeds up to 20 mph combat loaded and with rated towed load.
Gravel and dirt roads	1200 miles at varying speeds up to maximum combat loaded.
Gravel and dirt roads	400 miles at varying speeds up to maximum combat loaded and with rated towed load.
Cross country level and hilly terrain	1400 miles at varying speeds up to maximum combat loaded.
Cross country level and hilly terrain	200 miles at varying speeds up to maximum combat loaded and with rated towed load.

4.2.2.3.1 Test failure. Any failures or deficiencies of a workmanship or materials nature found during or as a result of the 4,000 mile test, will be cause for rejection of the vehicle. Further, the Government will refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of 4,000 mile test shall be prima facie evidence that all vehicles already accepted prior to completion of the 4,000 mile test are simi-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

larly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 Break-in run track. To determine conformance to 3.5.1, 3.5.2, and 3.5.3 each vehicle shall be lubricated and serviced prior to the break-in run. The run shall be on smooth, level, hard-surface roads, and for the distances specified in Table I. The vehicle shall be operated in reverse after each division as specified in 3.5.4. The undulating portion of the test shall be accomplished twice in each direction at speeds between 5 and 10 mph during division "a" and "b".

4.3.1 Simulated break-in run. To determine conformance to 3.5.1, 3.5.2 and 3.5.3, a simulated break-in run (see Table V) may be accomplished, in the skid type dynamometer in lieu of run specified in 4.3. Prior to the break-in run, each vehicle shall be warmed up in accordance with 3.5.2. Records shall be kept of the instruments readings. The final ten miles of the total break-in run shall be on smooth, level, hard surfaced roads. During this ten mile test, the vehicle shall be subjected to undulating operation, in accordance with 3.5.3 by being operated over the section of road conforming to figure 1. After completion of division "b" and "c" of Table V the vehicle shall be operated in reverse as specified in 3.5.4. The undulating portion of the test shall be accomplished twice in each direction and at speeds between 5 and 10 mph during division "d". The results of the break-in run in the first twenty vehicles, that meet all requirements, will serve as a standard for the evaluation of the balance of production.

Table V. Simulated break-in run

Division of run	Speed mph	Miles of test	Test conditions
a	0 to 10	10	Dynamometer
b	11 to 20	15	Dynamometer
c	21 to max.	5	Dynamometer
d	11 to 20	10	Track (road)

4.3.2 Condition after run. After the break-in run the vehicle shall be inspected to determine conformance to 3.5.5.

4.4 Inspection provisions.

4.4.1 Lot size. An inspection lot shall consist of all vehicles of one type, from one day's production, submitted at one time for acceptance inspection.

4.4.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

4.4.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table VI. The following constitutes a part of the classification of defects (see 6.6):

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any leak that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

Table VI. Classification of Defects

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
101. Engine	Malfunction, leaks (see 6.6) (b), mounting improper, damage	201. Improper component assembly or installation	Visual & functional
102. Engine governors	Malfunction, missing or defective seals	202. Improper setting	Functional
103. Power train	Malfunction, mounting improper, damage	203. Improper component assembly or installation	Visual & functional
104. Fuel system components	Malfunction, damage, leaks (see 6.6) (a)	204. Improper assembly, installation or clearance	Visual & functional
105. Fuel & oil lines valves & hose	Defective, clearance improper, malfunction, leaks (see 6.6) (a)	205. Improper assembly or installation	Visual & functional
106. Cooling system components engine & transmission	Malfunction, damage, leaks (see 6.6) (b)	206. Improper installation or assembly	Visual & functional
107. Exhaust system components	Damage, leaks	207. Improper assembly, installation or clearance	Visual
108. Electrical system components, lights (Includes wiring & instrumentation)	Malfunction, damage	208. Improper assembly, installation circuit identification, clearance or protection, illumination	Visual & functional

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

Table VI. Classification of Defects (Cont'd)

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
109. Suspension system components	Damage, tire or track pad separation, improper clearance or alinement, welding defects, leaks (see 6.6) (b) malfunction	209. Improper assembly, installation, bolt or nut torque and track tension	Visual

*NOTE: Filament failures are not to be classified as a major or minor defect.

110. Hull	Structures and welding defect		Visual
111. Controls & adjusting mechanisms	Malfunction, improper clearance	210. Improper assembly of installation, welding defects	Visual & functional
112. Doors, fenders, grilles, panels, ducts, stowage, brackets, boxes, racks & stops	Malfunction: torsion spring and hold-open devices	211. Improper fits, adjustment assembly or installation	Visual & functional
113. Drain valves	Valve seating improper	212. Improper installation, assembly or control adjustment	Visual & functional
114. Cushions, seats, rests & crash pads	Seat control malfunction	213. Improper fit, installation or adhesion damage	Visual & functional
115. Boom	Malfunction, damage	214. Improper control, adjustment	Visual & functional
116. Winches	Malfunction, damage	215. Improper control, adjustment	Visual & functional

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

Table VI. Classification of Defects (Cont'd)

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
117. Spade & controls	Malfunction, mis-alinement	216. Improper assembly or installation	Visual & functional
118. Traversing mechanism	Clearance improper, component damage	217. Improper assembly, installation or chain adjustment	Visual & functional
119. Hydraulic power pack, motor & lines	Component damage, leaks (see 6.6 (b)	218. Improper assembly, installation hydraulic fluid level & type improper	Visual & functional
120. Misc. items or assessories	Malfunction, damage, missing	219. Improper assembly or installation	Visual & functional
121. Fire extinguisher system components	Seals missing or defective	220. Data card incomplete, improper installation or assembly	Visual
122. Speed	Non-conformance		Functional
123. Acceleration	Non-conformance		Functional
124. Turning ability & steering	Non-conformance, wander, side-pull		Functional
125. Braking ability (parking & service brakes	Non-conformance: grade holding or stopping distance, equalization improper		Functional

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

Table VI. Classification of Defects (Cont'd)

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
Paint protective & anti-skid coatings		221. Application, coverage & materials improper	Visual
Lube fittings & lubrication		222. Fittings missing, defective or improperly installed, lubrication improper	Visual
Decals, markings data & instruction plates		223. Incomplete data, missing, improper location or size, illegible	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

4.4.2.1 Unclassified defects. All defects having no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty-day period will be added to the minor classification with no increase in AQL's.

4.4.2.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found (see 6.6).

4.4.2.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. Correction of all deficiencies shall be accomplished prior to acceptance (see 6.6).

4.5 Classification of tests.

(a) Acceptance tests 4.5.1 through 4.5.1.27.
(b) Control tests 4.5.2 through 4.5.2.9.
(c) Inspection comparison tests (ICT) 4.5.1 through 4.5.3.15.

4.5.1 Acceptance tests. To determine conformance to 3.6, after break-in run, (see 3.5) each vehicle shall be operated for a distance of eight miles or more by the contractor subjected to acceptance tests specified in Table VIII. Except for equipment and compenent installation and servicing performed by the contractro as a final processing of vehicle for acceptance, vehicles shall be completely assembled and serviced with fuel tanks filled to capacity, lubricants and fluid as specified, but need not be combat loaded except for holding test as specified in 4.5.1.19. After the test, the vehicle shall be examined for evidence of fuel or lubricant leakage or other deficiencies.

4.5.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance until evidence has been provided by the contractor that the corrective action has been taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

Table VIII. Location for tests

Title	Paragraph Numbers	Place of Manufacturer	Government Proving Ground
Acceptance tests	4.5.1	X	X
Sealing and seals leakage check	4.5.1.2	X	X
Sealing and seals check	4.5.1.2.1	X	X
Hatch and door seal check	4.5.1.2.2	X	X
Fuel system slope check	4.5.1.3	X	X
Hydraulic reservoir check	4.5.1.4	X	X
Suspension lockout test	4.5.1.5	X	X
Controls and adjustment mechanisms	4.5.1.6	X	X
Throttle linkage check	4.5.1.7	X	X
Fuel shut-off valve check	4.5.1.8	X	X
Seats check	4.5.1.9	X	X
Generator voltage check	4.5.1.10	X	X
Light check	4.5.1.11	X	X
Slip ring (hull to turret) and all communication circuits check	4.5.1.12	X	X
Vision devices and receptacle check	4.5.1.13	X	X
Power package and train test	4.5.1.14	X	X
Speed test	4.5.1.15	X	X
Acceleration test	4.5.1.16	X	X
Engine starting on grades and slopes test	4.5.1.17	X	X
Stopping test	4.5.1.18	X	X
Holding test	4.5.1.19	X	X
Turning test	4.5.1.20	X	X
Shallow water fording test	4.5.1.21	X	X
Main hydraulic system check	4.5.1.22	X	X
Descent rate of boom and boom test	4.5.1.23	X	X
Winch tests	4.5.1.24	X	X
Winch brake test	4.5.1.25	X	X
Spade test	4.5.1.26	X	X
Traversing mechanism tests	4.5.1.27	X	X
Control tests	4.5.2	X	X
Contamination test	4.5.2.3	X	X
Carbon monoxide concentration test	4.5.2.4	X	X
Fuel tank test	4.5.2.5	X	X
Fasteners and lock checks	4.5.2.6	X	X
Stowed equipment test	4.5.2.7	X	X
Engine air induction system test	4.5.2.8	X	X
Fire extinguisher test	4.5.2.9	X	X
Comparison test	4.5.3		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

Table VIII. Location for tests (Cont'd)

Title	Paragraph Numbers	Place of Manufacturer	Government Proving Ground
Winterization equipment test	4.5.3.2		X
Air cleaner check	4.5.3.3		X
Fuel system operational test	4.5.3.4		X
Communication equipment test	4.5.3.5		X
Environmental tests	4.5.3.6		X
Cooling system tests	4.5.3.7		X
Speed test with towed load	4.5.3.8		X
Climbing test	4.5.3.9		X
Trench crossing tests	4.5.3.10		X
Vertical obstacle tests	4.5.3.11		X
Hydraulic endurance operational test	4.5.3.12		X
Boom, winch and spade test	4.5.3.12.1		X
Traversing tests	4.5.3.12.2		X
Emergency hydraulic pump check	4.5.3.12.3		X
Level wind device test	4.5.3.12.4		X
Electromagnetic compatibility test	4.5.3.13		X
Vibration and noise level check	4.5.3.14		X
Sound level meter and microphone	4.5.3.14.1		X
Fire extinguisher system operational test	4.5.3.15		X

4.5.12 Sealing and seals leakage check.

4.5.1.2.1 Sealing and seals check. To determine conformance to 3.4.11 and 3.4.12, prior to and after fording operations the vehicle shall be checked for the amount of water in vehicle, and leaking of lubricants.

4.5.1.2.2 Hatch and door seal check. To determine conformance to 3.4.1.3, the hatches and doors shall be closed and locked, and a spray of water directed on each hatch and door for a period of 3 minutes. All hatches and doors shall be checked for leakage.

4.5.1.3 Fuel system slope check. To determine conformance to 3.4.5, 3.4.5.1, and 3.4.5.2 with the vehicle on specified grades and slopes; the engine operating between 750 and 1000 rpm the fuel system shall be checked for fuel leaks and fuel supply to engine.

4.5.1.4 Hydraulic reservoir check. To determine conformance to 3.4.6, prior to filling the hydraulic reservoir shall be checked for cleanliness and leaks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

4.5.1.5 Suspension lockout test. To determine conformance to 3.4.7, the suspension lockout shall be operated and checked for functional requirements.

4.5.1.6 Controls and adjustment mechanisms check. To determine conformance to 3.4.8 and 3.4.9, all controls shall be operated and checked for functional requirements. Adjustment mechanisms shall be examined for proper adjustment.

4.5.1.7 Throttle linkage check. To determine conformance to 3.4.13, the throttle pedal shall be operated to full throttle, and check the engine throttle rod to ensure that full rack position has been reached.

4.5.1.8 Fuel shut-off valve check. To determine conformance to 3.4.14, with the engine operating at 750 to 900 rpm, the fuel shut-off handle shall be pulled to the "off" position. The time for the engine to stop operation shall be recorded.

4.5.1.9 Seats check. To determine conformance to 3.4.15, the seats shall be operated and checked for functional requirements.

4.5.1.10 Generator voltage check. To determine conformance to 3.4.16.1, the generator voltage shall be checked using a calibrated master voltage meter.

4.5.1.11 Light check. To determine conformance to 3.4.16.2, all lights shall be operated during and after the break-in run throughout all stages of operation, and checked for functional requirements.

4.5.1.12 Slip ring (hull to turret) and all communication circuits check. To determine conformance to 3.4.16.3, the slip ring (hull to turret) and all communication circuits of each vehicle shall be checked for continuity and shorts.

4.5.1.13 Vision devices and receptacle check. To determine conformance to 3.4.17, the vision devices installed in their applicable receptacle shall be sprayed with water directed on each devise for a period of 3 minutes. All devices shall be checked for leakage and functional requirements.

4.5.1.14 Power package and train test. To determine conformance to 3.6.2, the vehicle shall be operated throughout all gear and speed ranges, the power plant, power train, final drives, tracks, and suspension system, and applicable controls checked for functional requirements.

4.5.1.15 Speed tests. To determine conformance to 3.6.3.1, vehicle shall be operated at specified speeds and checked for functional requirements.

4.5.1.16 Acceleration test. To determine conformance to 3.6.4, the vehicle shall be accelerated from a standing start to the required miles per hour, and the time recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

4.5.1.17 Engine starting on grades and slopes test. To determine conformance to 3.5.2 and 3.6.5 the engine shall start, oil pressure and temperature shall be maintained when operating on specified grades and slopes. The 60 percent grade test may be conducted in conjunction with brake holding test.

4.5.1.18 Stopping test. To determine conformance to 3.6.6.1, the vehicle shall be operated on dry level hard-surfaced road without loose material, and checked for functional requirements.

4.5.1.19 Holding test. To determine conformance to 3.6.6.2, the vehicle shall be driven up the specified grade, service brakes applied and held for not less than one minute and vehicle observed for movement. With brakes locked for parking the vehicle shall be held stationary for not less than 2 minutes and observed for movement. During this test vehicle shall be combat loaded or loaded with a simulated load of equal weight shall be placed in a position that will not restrict the flow of engine air (intake of exhaust).

4.5.1.20 Turning test. To determine conformance to 3.6.7, the vehicle shall be operated as specified in full 360 degree pivot turns and checked for functional requirements.

4.5.1.21 Shallow water fording test. To determine conformance to 3.6.8, the vehicle shall be driven into water of specified depth, without special equipment. Water accumulation shall not exceed specified depth in hull bottom after 15 minutes of fording operation. The engine shall idle at 1000 rpm for 15 minutes, be stopped, and shall restart in not more than 3 minutes with all accessories functioning satisfactorily during and after fording.

4.5.1.22 Main hydraulic system check. To determine conformance to 3.6.12, the main hydraulic system shall be operated throughout all ranges of operation and checked for meeting performance requirements.

4.5.1.23 Descent rate of boom and boom test. To determine conformance to 3.6.12.1 and 3.6.12.2, the boom shall be operated as specified and checked for meeting performance requirements.

4.5.1.24 Winch tests. To determine conformance to 3.6.12.3 through 3.6.12.5, the winches shall be operated as specified, with intended loads or a simulated load of equal weight and checked for performance requirements.

4.5.1.25 Winch brake test. To determine conformance to 3.6.12.5 with winches operating under specified loads, the brakes shall be operated and observed for stopping and holding maximum loads.

4.5.1.26 Spade test. To determine conformance to 3.6.12.7, the spade shall be operated through all ranges of operation and checked for functional requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

4.5.1.27 Traversing mechansim test. To determine conformance to 3.6.12.8 through 3.6.12.8.3, the traversing mechanism shall be operated throughout entire ranges of operation and checked for functional requirements.

4.5.2 Control tests. Control tests for maintaining control of those requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein.

4.5.2.1 Frequency. The first vehicle off the production line, or one of the first ten vehicles produced, and thereafter one per month, shall be checked in accordance with the control test specified herein.

4.5.2.2 Failure. If the vehicle selected fails to pass any of the control tests the Government inspector shall stop acceptance examination and testing on subsequent lots until evidence has been provided by the contractor that corrective action has been taken.

4.5.2.3 Contamination test. To determine conformance to 3.4.1.2 the vehicle after the water fording test, shall be examined for contamination of lubricants of the transmission, engine final drive, all road wheels and one of their corresponding arms, on each side of the vehicle. Contamination shall not exceed 2 percent.

4.5.2.4 Carbon monoxide concentration test. To determine conformance to 3.4.3 the crew compartments shall be checked for carbon monoxide concentration.

4.5.2.5 Fuel tank test. To determine conformance to 3.4.5.1, the fuel tanks shall be filled to capacity at a rate of not less than 50 gallons per minute. The tanks and lines shall be checked for leaks before and after a two mile road test on smooth, level, hard-surfaced roads.

4.5.2.6 Fasteners and lock checks. To determine conformance to 3.4.10, all fasteners and locks shall be operated and checked for functional requirements.

4.5.2.8 Stowed equipment test. To determine conformance to 3.4.11, all manufacturer and depot installed OVE shall be stowed (combat loaded) on the selected vehicle. All items shall fit with proper clearance in the spaces provided. The equipment shall be removed after the test. The OVE, used shall be of the latest production available to the contractor; and the contractor shall be permitted to unpackage and repackage (for domestic shipment only) equipment used for test.

4.5.2.8 Engine air induction system test. To determine conformance to 3.4.12, the engine air induction system shall be checked for internal cleanliness and for specified pressure.

4.5.2.9 Fire extinguisher test. To determine conformance to 3.10, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

fire extinguisher system shall be functionally tested. The first vehicle tested shall have the fire extinguisher discharged by the external and internal controls and the discharge force recorded.

4.5.3 Comparison tests. The Government may select vehicles any time during the contract production period, and subject vehicle to tests listed in Table VIII and Table IX to reveal deficiencies of manufacture that may reduce the effective operation of these items in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving ground designated by the contracting officer. Selection of vehicles shall be on a spot check basis. Comparison test vehicles shall be combat loaded with all on-vehicle equipment stowed in their applicable spaces to conform to 3.4.11. Vehicles selected shall not include any vehicles previously tested for conformance to 4.5.2 (control tests).

4.5.3.1 Test failure. Failure of any vehicle tested as specified in 4.5.3 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacture that corrective action has been taken to eliminate the conditions.

Table IX. 2000 Mile test combat loaded
(Government Proving Grounds)

Course	Mileage & Speeds
Hard surface roads	300 miles at varying speeds up to 34 mph
Gravel and dirt roads	1300 miles at varying speeds up to maximum
Hilly cross country	200 miles at varying speeds up to maximum
Level cross country	200 miles at varying speeds up to maximum

4.5.3.2 Winterization equipment test. To determine conformance to 3.4.2, the vehicle properly serviced and equipped, shall be subjected to specified ambient temperatures, and observed for satisfactory starting and operation of the engine and the function personnel heaters.

4.5.3.3 Air cleaner check. To determine conformance to 3.4.4, during and after operation, the air cleaners shall be checked for functional requirements.

4.5.3.4 Fuel system operational test. To determine conformance to 3.4.5, 3.4.5.1 and 3.6 in conjunction with all vehicle slope tests, the vehicle fuel system shall be checked for functional requirements. The tanks shall be filled as specified and rate of filling recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

4.5.3.5 Communications equipment test. To determine conformance to 3.4.16.3 communication system equipment shall be installed, subjected to tests and checked for voltage supply.

4.5.3.6 Environmental tests. To determine conformance to 3.6.1, the vehicle, properly serviced and equipped, shall be subjected to the applicable tests in high and low temperatures.

4.5.3.7 Cooling system tests. To determine conformance to 3.6.2.1 and 3.6.2.1.1 the vehicle shall be operated, as specified, and cooling system checked for functional requirements.

4.5.3.8 Speed test with towed load. To determine conformance to 3.6.3.2, each vehicle selected shall be operated, as specified, and checked for meeting performance requirements.

4.5.3.9 Climbing test. To determine conformance to 3.6.9, the vehicle shall be driven in forward and reverse gear up specified grade.

4.5.3.10 Trench crossing tests. To determine conformance to 3.6.10, the vehicle shall be driven over trenches, as specified, without stalling and damage to the vehicle or components.

4.5.3.11 Vertical obstacle tests. To determine conformance to 3.6.11, the vehicle shall be driven over trenches, as specified, without stalling and damage to the vehicle or components.

4.5.3.12 Hydraulic system endurance operational test. To determine conformance to 3.2.1.2 and 3.6.12 the system shall be operated as specified and checked for meeting performance requirements. During these tests an auxiliary cooling device may be employed to prevent a reservoir oil temperature in excess of 170°F, to provide for continuous hydraulic test operation.

4.5.3.12.1 Boom, winch, and spade test. To determine conformance to 3.2.1.2, 3.6.12.1 through 3.6.12.7, the following main winch endurance test shall be performed. Emplace the vehicle spade and attach the main winch cable to the test load. Pay-out a minimum of 100 feet of cable under a load of 20,000 pounds. Using a test load representing 90 percent of rated full drum line pull, pay-in winch cable, then return engine and hydraulic system to idle and pause eight minutes. Repeat the above duty cycle until a total of eight hours of operation has been accumulated. Only the time required to conduct the entire pay-out, pay-in cycle shall be counted toward the eight hours of operation. Pay-in line speed shall be maintained at rated speed.

4.5.3.12.2 Traversing tests. To determine conformance to 3.6.12.8 and 3.6.12.8.1, all traversing mechanisms shall be tested as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

(a) Hoist and lower a 30,000 pound load pausing fifteen seconds between lifts. The load shall be raised to its maximum height and then alternately lowered to within the foot of the ground. Repeat the above duty cycle until four hours of operation have been completed.

(b) With the 30,000 pound load directly behind the vehicle, lift the load to its maximum height, then lower to within one foot of the ground, then traverse the load 90 degrees toward the operator's left, then traverse the load 180 degrees toward the operator's right, then traverse the load 90 degrees back to the starting position, then set the load on the ground and pause fifteen seconds. Repeat the above duty cycle until four hours of operation has been accumulated.

(c) In both of the above test phases the hoisting line speed shall be maintained at full rated capacity and lowering line speed shall be maintained at approximately one-half rated capacity. Also, the pauses between lifts will not be credited to the eight hours of endurance specified.

4.5.3.12.3 Emergency hydraulic pump check. To detemermine conformance to 3.6.12.8.2, the emergency hydraulic pump shall be operated and checked for functional requirements.

4.5.3.12.4 Level wind device test. To determine conformance to 3.6.12.8.3, winches shall be operated as specified and the level wind device observed for meeting performance requirements.

4.5.3.13 Electromagnetic compatibility test. To determine conformance to 3.7, the vehicle shall be tested in accordance with the applicable provisions of MIL-STD-461 (see 6.8).

4.5.3.14 Vibration and noise level check. To determine conformance to 3.9 through 3.9.3, the average of 3 noise level readings at each octave band shall be determined for the following situations and combinations.

(a) At each operation head position under normal operational procedures.
(b) At each instruction and trainee head position where training is conducted.
(c) At each maintenance personnel head position, where maintenance is conducted with the equipment operating.
(d) All positions where other personnel may be present in or on the equipment, representative sample readings shall be made above.

4.5.3.14.1 Sound level meter and microphone. The sound level meter and microphone used shall be in accordance with applicable documents (see 6.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

4.5.3.15 Fire extinguisher system test. To determine conformance to 3.10, the fire extinguisher system shall be separately exercised by internal and external controls and results examined for distribution and discharge requirements.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage shall be in accordance with MIL-R-4671C

6. NOTES

6.1 Intended use. The vehicle is intended for use by the Armed Forces as a recovery vehicle up to 30 ton loads and servicing of all vehicles.

6.2 Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction model if required (see 3.1).
(c) Special kits if required (see 3.4.2).
(d) Radio suppression if other than specified (see 3.7).
(e) Painting if other than specified (see 3.8.1).
(f) Markings if other than specified (see 3.8.2).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

6.4 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specified in the classifications of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process average} \quad \frac{\text{Number of Defects}}{\text{Number of Vehicles Inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

6.5 The following shall be used with the classification of defects:

(a) Weep: - Any evidence of fluid beyond the seal.
(b) Seep: - Any evidence of fluid beyond the seal that does not result in function of a droplet.
(c) Droplet: - Any evidence of fluid beyond the seal that results in the formation of a droplet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

(d) Drip: - Any evidence of fluid beyond the seal where droplet forms and falls.

6.6 Definitions of recurring major and minor deficiencies. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspection will assure that all deficiencies encountered during the inspection are enumerated on the deficiency sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.7 The following publications should be used in determining sound pressure level.

ADA Z24.10-1953	Octave-Band Filter Set for the Analysis of Noise and Other Sounds
ASA S1.1-1960	Acoustical Terminology
ASA S1.4-1961	General Purpose Sound Level Meters

(Applications for copies of these documents should be addressed to: American Standards Association, 10 East 40th St., New York 16, N. Y.)

6.8 Electromagnetic compatibility test. The contracting officer should submit a copy of the report(s) required in 4.5.3.13 to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey 07703, for approval.

6.9 International standardization agreement. Certain provisions (3.6.8) of this specification are the subject of international standardization agreement NATO STANAG 2805-A (Edition 3). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Custodian:
Army - MO

Preparing activity:
Army - MO (ATAC)

Project No. 2320-A716

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.

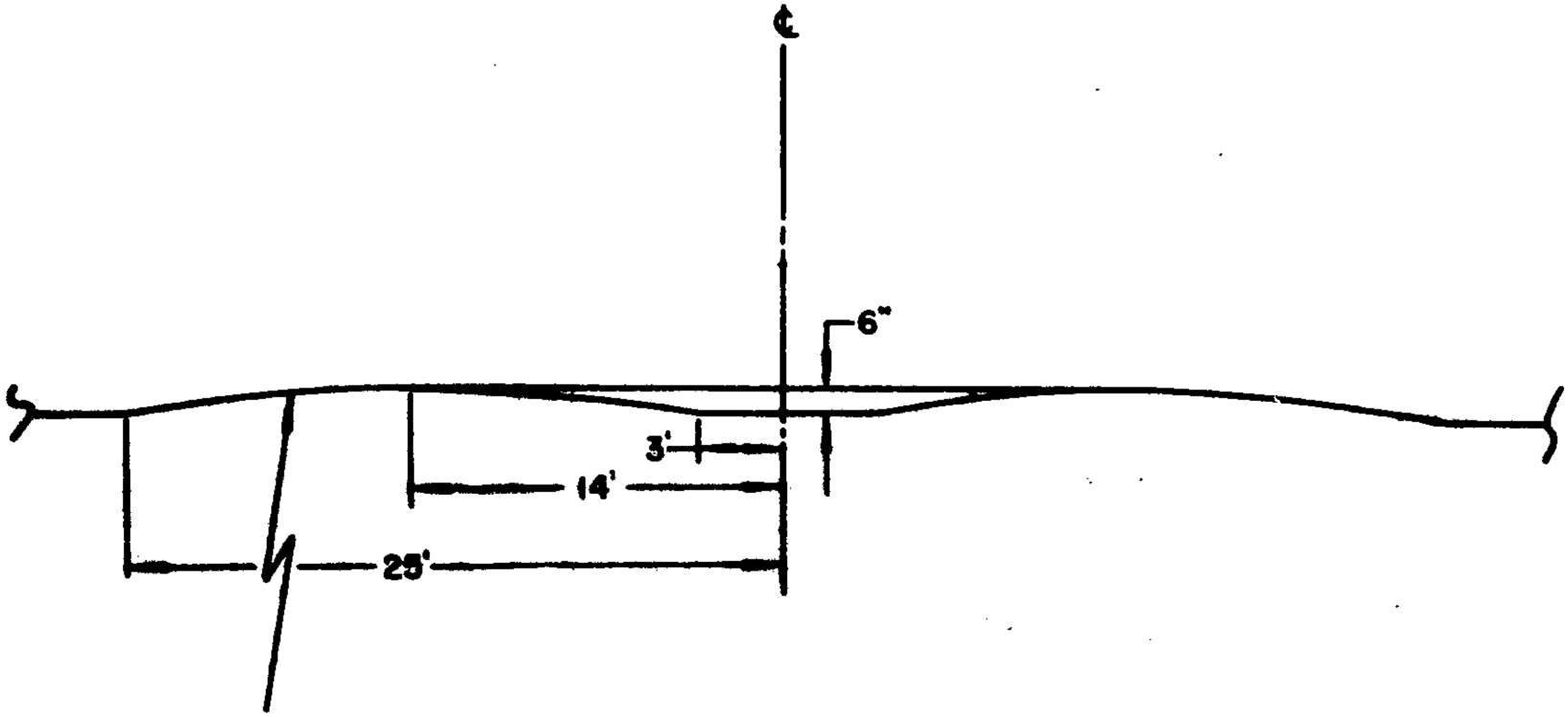


FIGURE 1 - PROFILE OF TEST TRACK
(ARTIFICIAL BUMP)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:45Z
Check the source to verify that this is the current version before use.